Risk Across Domains:

Alignment and Rebalancing in CEOs Risk Allocation

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ABSTRACT

How do CEOs allocate risk across the personal and corporate domains in which they operate? Prior research has largely examined how differences in managerial risk preferences translate into firm-level outcomes, often treating corporate decisions as direct expressions of underlying preferences. We propose a complementary perspective in which CEOs manage total risk exposure across domains, with personal financial portfolios and corporate policies jointly contributing to that exposure. This perspective highlights a central tension between cross-domain alignment and intertemporal rebalancing in CEO risk-taking, yielding two related predictions. First, across CEOs, higher personal financial risk exposure is associated with riskier corporate financial policies, reflecting differences in target exposure. Second, within a given CEO over time, changes in personal portfolio risk induce compensatory adjustments in firm-level risk-taking, as CEOs rebalance toward their target level of exposure. We test these predictions using detailed longitudinal data on CEOs' personal financial portfolios and firm-level financial policies in Norway. Consistent with our framework, we find that CEOs with higher personal risk exposure pursue higher leverage and lower cash holdings. At the same time, within CEO–firm matches, increases in personal portfolio risk are associated with reductions in firm-level risk-taking, and vice versa. These relationships are stronger when CEOs have greater power relative to the board. Our findings introduce a cross-domain portfolio perspective on managerial decision-making and show that cross-sectional alignment and intertemporal rebalancing can be understood as complementary manifestations of a common underlying mechanism of risk allocation.

Keywords: CEO risk taking, risk allocation, corporate financial policy, managerial discretion, capital structure, upper echelons theory

INTRODUCTION

Research on executive decision-making has long emphasized that CEOs differ systematically in their risk preferences and that these differences shape firm-level outcomes (Hambrick and Mason 1984; Finkelstein, Hambrick, and Cannella 2009). A large body of work shows that more risk-tolerant CEOs tend to pursue riskier corporate policies, including higher leverage, more acquisitive strategies, and greater exploration (Chatterjee and Hambrick 2007; Hirshleifer, Low, and Teoh 2012). More recent work further highlights that such individual-level heterogeneity can translate into systematic variation in firms' strategic choices, even when firms face similar external conditions (Boustanifar, Zajac, and Zilja 2022).

This literature has primarily focused on how differences across CEOs translate into differences across firms, often treating firm-level decisions as direct expressions of underlying preferences. However, this perspective largely treats risk-taking within the firm as an isolated expression of preferences, without considering how CEOs simultaneously manage risk across multiple domains of their lives. In this study, we revisit this perspective by considering that CEOs make risk-related decisions across multiple domains, most notably in their personal financial portfolios and in the firms they lead. This raises a fundamental question: Do CEOs express their risk preferences independently within each domain, or do they allocate risk across domains in a way that reflects an overall target level of exposure, giving rise to cross-domain alignment and intertemporal rebalancing?

We argue that both cross-domain alignment and intertemporal rebalancing arise from a common underlying mechanism of cross-domain risk allocation. Specifically, rather than viewing risk-taking as the independent expression of preferences within separate domains, we conceptualize it as a process in which CEOs allocate risk exposure across domains to

maintain a target level of total exposure. In doing so, we shift the focus from how CEO traits are expressed within domains to how risk is *distributed across domains*, introducing a new lens on managerial decision-making.

This perspective yields two distinct but related predictions. First, across CEOs, those with higher target risk exposure will tend to pursue riskier corporate financial policies, generating the familiar pattern of cross-sectional alignment between personal and firm-level risk-taking. Second, within a given CEO over time, changes in personal portfolio risk driven by asset value fluctuations can induce compensatory adjustments in firm-level risk-taking, as CEOs rebalance toward their target level of exposure.

This second prediction highlights a dynamic, within-CEO adjustment process that has received limited attention in prior work. In particular, it suggests that increases in the riskiness of a CEO's personal financial portfolio—driven by changes in asset values—may lead to decreases in firm-level financial risk, and vice versa. By integrating these cross-sectional and intertemporal dynamics, our framework offers a unified account of when CEO risk preferences manifest as alignment across domains and when they generate offsetting adjustments across domains.

We test these ideas using detailed longitudinal data on CEOs' personal financial portfolios and firm-level financial policies in Norway. Our data allow us to construct a revealed-preference measure of CEO risk exposure based on the composition of their personal financial assets and to examine how both cross-sectional differences and within-CEO changes in portfolio risk relate to firm leverage and cash policies. Consistent with our arguments, we find that CEOs with higher personal risk exposure tend to pursue riskier corporate financial policies across firms, but that within a given CEO–firm match, increases in personal portfolio risk are associated with reductions in firm-level risk-taking. These relationships are amplified when CEOs have greater power relative to the board.

This study contributes to research on managerial risk-taking and corporate decision-making in three main ways. First, we introduce a *cross-domain portfolio perspective* on CEO risk-taking, highlighting that personal and professional decisions are jointly determined rather than independently expressed. We propose that CEOs allocate risk exposure across domains to maintain a target level of total exposure, adjusting decisions in one domain in response to changes in another. In doing so, we complement prior work that links CEO-level traits—such as risk propensity—to firm-level outcomes (e.g., Boustanifar, Zajac, and Zilja (2022)) by shifting attention from trait expression within a domain to the allocation of risk across domains.

Second, we reconcile two seemingly opposing empirical patterns—cross-sectional alignment and intertemporal rebalancing—which we treat as a central theoretical tension in CEO risk-taking, by showing that both can arise from a common underlying mechanism of target risk allocation. Third, we provide new empirical evidence based on revealed personal financial behavior, offering a more direct and behaviorally grounded measure of managerial risk exposure and addressing longstanding challenges in measuring managerial risk preferences (Heaton and Lucas 2000; Calvet, Campbell, and Sodini 2009).

THEORY AND HYPOTHESES

CEO Risk-Taking as Cross-Domain Risk Allocation

Research on managerial risk-taking has long emphasized that executives differ in their willingness to take risks and that such differences shape organizational outcomes (Hambrick and Mason 1984; Carpenter, Geletkanycz, and Sanders 2004). Consistent with this perspective, prior research shows that variation in CEO risk propensity can lead to systematic differences in firms' strategic choices (Boustanifar, Zajac, and Zilja 2022). This literature has

typically treated observed decisions—such as capital structure choices, acquisitions, or innovation investments—as direct expressions of underlying risk preferences. Implicit in this approach is the assumption that preferences are enacted within a given domain, with less attention paid to how decisions in one domain may relate to decisions in others (Schoemaker 1990).

In this paper, we shift from a within-domain perspective to a cross-domain view of CEO risk-taking. To do so, we distinguish between two related but conceptually distinct constructs: *risk appetite* and *risk exposure*. Risk appetite refers to an individual's latent preference for engaging in risky activities in pursuit of valued outcomes (Weber, Blais, and Betz 2002; Dohmen et al. 2011). Risk exposure, in contrast, reflects the realized level of risk embedded in an individual's portfolio of decisions (Thaler and Johnson 1990; Gollier 2001). Risk appetite is often assumed to be relatively stable over time, although it may evolve with experience and context. Building on this distinction, we conceptualize CEO risk-taking as a cross-domain allocation problem.

We propose that CEOs do not simply express risk preferences within domains but instead allocate risk across domains to maintain a target level of total exposure. CEOs simultaneously make risk-related decisions in multiple domains, most notably in their personal financial portfolios and in the firms they lead. Rather than viewing these decisions as independent expressions of preferences, we propose that they are jointly determined as components of a broader allocation of risk exposure across domains. Under this view, observed choices in any single domain reflect not only underlying preferences but also the current configuration of risk exposure in other domains.

This perspective implies that CEOs seek to maintain a target level of total risk exposure across domains. While this target is shaped by underlying risk appetite, the realized level of exposure may deviate from it due to changes in asset values, constraints on

adjustment, and other contextual factors. As a result, CEO decision-making involves not only the expression of preferences within domains but also the adjustment of exposure across domains. Importantly, our argument does not require that risk appetite itself changes over time, but rather that realized risk exposure may deviate from a target level and be adjusted across domains.

We also do not assume that CEOs explicitly compute or continuously monitor a precise optimal level of total risk exposure. Rather, the notion of a 'target' should be understood as a reduced-form representation of a preferred range of exposure shaped by preferences, experience, and constraints. Deviations from this range can trigger adjustments across domains, even in the absence of deliberate optimization. Consistent with prior work on adaptive risk-taking, such adjustments may arise from shifts in perceived overall exposure or comfort with risk, rather than from explicit portfolio optimization, which in turn influence subsequent risk-taking choices (Sitkin and Pablo 1992; Thaler and Johnson 1990).

This cross-domain perspective implies two distinct but related empirical predictions. First, across CEOs, those with higher target risk exposure will tend to pursue riskier corporate financial policies, generating cross-sectional alignment between personal and firm-level risk-taking. Second, within a given CEO over time, changes in personal portfolio risk—driven by asset value fluctuations—can induce compensatory adjustments in firm-level risk-taking, as CEOs rebalance toward their target level of exposure.

The first prediction reflects cross-sectional differences in target risk exposure across CEOs, whereas the second captures within-CEO adjustments in response to deviations from that target. Together, these predictions provide a unified account of when CEO risk-taking manifests as alignment across domains and when it generates offsetting adjustments across domains. Importantly, these two empirical patterns are not independent; both reflect adjustments toward a common underlying level of total risk exposure across domains. More

broadly, our framework formalizes the tension between cross-domain alignment and intertemporal rebalancing as a central organizing feature of CEO risk-taking.

Cross-Sectional Implications: Baseline Alignment Across Domains

If CEOs differ in their target levels of total risk exposure, then these differences should manifest in both personal and corporate domains. CEOs with higher target risk exposure will allocate a greater proportion of their personal wealth to risky assets and will also tend to pursue riskier corporate financial policies, such as higher leverage and lower cash holdings. This prediction is consistent with prior research documenting that managerial characteristics related to risk tolerance are associated with firm-level risk-taking (Malmendier and Tate 2005; Cronqvist, Makhija, and Yonker 2012). Within our framework, however, this pattern reflects not simply cross-situational consistency, but a common underlying target level of exposure that is expressed across domains. This logic yields two complementary implications: one reflecting differences in target exposure across CEOs, and the other reflecting adjustments toward that target within CEOs over time. Formally, we state the following baseline hypothesis:

> **Hypothesis 1 (Baseline alignment):** CEOs with higher personal financial risk exposure will be associated with higher firm-level financial risk-taking.

Intertemporal Rebalancing: Adjustments Toward Target Risk Exposure

While cross-sectional differences reflect stable differences in target risk exposure, realized risk exposure may deviate from these targets over time. In particular, changes in the value of assets held in a CEO's personal financial portfolio—especially those driven by market movements rather than active trading—can generate variation in personal risk

exposure that is less directly tied to contemporaneous firm-level decision-making (Campbell 2006; Calvet, Campbell, and Sodini 2009).

We argue that such changes alter the CEO's overall risk exposure across domains. For example, an increase in the relative value of risky assets raises the riskiness of the CEO's personal portfolio, increasing total exposure, whereas a decrease reduces overall exposure. These changes need not reflect deliberate choice but instead arise mechanically from asset price fluctuations.

When such shifts occur, CEOs may adjust risk-taking in other domains in ways that move overall exposure back toward a preferred range. In particular, increases in personal portfolio risk are expected to be associated with reductions in firm-level financial risk-taking, whereas decreases in personal risk are expected to be associated with increases in firm-level risk-taking. This pattern reflects intertemporal rebalancing across domains rather than consistency within a single domain.

Importantly, this argument does not require that CEOs explicitly compute or continuously monitor an optimal allocation of risk across domains. Rather, it is consistent with the broader idea that individuals respond to changes in their risk environment in ways that move them toward a preferred level or range of exposure, whether consciously or unconsciously (Thaler and Johnson 1990). Whereas cross-sectional differences reflect variation in target risk exposure across CEOs, realized exposure for a given CEO may deviate from that target over time. Formally, we state:

> **Hypothesis 2 (Rebalancing):** Within a given CEO–firm match, increases in the riskiness of a CEO's personal financial portfolio will be associated with decreases in firm-level financial risk-taking, whereas decreases in the riskiness of a CEO's personal financial portfolio will be associated with increases in firm-level financial risk-taking.

Why Adjust the Firm Rather Than the Personal Portfolio?

A natural question is why CEOs would adjust firm-level policies rather than simply rebalance their personal portfolios. While personal portfolio adjustment is one potential response, several factors suggest that firm-level decisions may serve as an important margin of adjustment.

CEOs may face constraints on personal trading, including regulatory restrictions, trading windows, tax considerations, and illiquidity of certain holdings, which can limit the speed or extent of personal rebalancing. Consistent with this view, prior evidence suggests that individual portfolios often exhibit substantial inertia and infrequent rebalancing, implying that adjustments through other domains may play a nontrivial role (Agnew, Balduzzi, and Sunden 2003; Calvet, Campbell, and Sodini 2009). Indeed, consistent with this evidence, our data also show that managers trade infrequently: the median number of trades per year is zero, and the average is three. At the same time, firm-level financial policies—such as leverage and cash management—are continuously adjustable and fall directly under the CEO's decision authority, making them salient and controllable levers for influencing overall risk exposure. Moreover, adjustments at the firm level may allow for larger or more immediate changes in total exposure than marginal adjustments in personal portfolios.

Taken together, these considerations suggest that firm-level decisions can serve as one important margin of adjustment when overall exposure shifts. Importantly, we do not assume that firm-level adjustments fully offset changes in personal portfolio risk or that they are the only margin of adjustment; rather, they represent one channel through which CEOs may respond to changes in their overall risk exposure, particularly when personal portfolio adjustments are constrained or costly. Consistent with this interpretation, firm-level and

personal adjustments may operate jointly as complementary margins of adjustment across domains.

Alternative Explanations and Theoretical Distinctions

The predictions above may appear related to other well-established mechanisms in the literature. We consider two related alternative explanations. First, prospect theory suggests that individuals become more risk-averse in the domain of gains and more risk-seeking in the domain of losses (Kahneman and Tversky 1979). While changes in personal wealth may influence risk-taking through such mechanisms, our framework differs in two important respects. Prospect theory predicts changes in risk-taking within a given domain as a function of gains and losses, whereas our framework predicts systematic adjustments across domains in response to changes in overall risk exposure. As such, prospect theory does not imply cross-domain substitution between personal and corporate risk-taking, nor does it predict the symmetric adjustments we observe in response to both increases and decreases in exposure.

Second, prior work has argued that managers adjust firm policies to diversify their exposure to firm-specific risk (e.g., Amihud and Lev (1981)). While related, our framework is broader in scope. Rather than focusing on the concentration of wealth in the focal firm, we consider changes in overall risk exposure, including those driven by assets outside the firm. Accordingly, our framework predicts adjustments in firm-level risk-taking in response to shifts in total exposure, regardless of whether those shifts originate within or outside the firm, and thus extends beyond explanations based solely on firm-specific risk diversification. Taken together, these distinctions highlight that our framework captures a cross-domain allocation mechanism that is not implied by existing within-domain or firm-specific explanations.

CEO Power and the Implementation of Risk Allocation

If CEOs actively allocate risk exposure across domains, then their ability to implement such allocation should depend on their discretion within the firm. When CEOs have greater power relative to the board, they are better able to translate their preferred allocation of risk exposure into firm-level financial policies (Finkelstein 1992; Boustanifar, Zajac, and Zilja 2022). Accordingly, both the cross-sectional alignment and intertemporal rebalancing relationships described above should be stronger when CEOs have greater power within the firm. Formally, we state:

> **Hypothesis 3 (Moderation by CEO power):** The relationships predicted in Hypotheses 1 and 2 will be stronger when CEOs have greater power relative to the board.

METHODS

Data and Sample

Our empirical setting is Norway, where detailed administrative data allow us to observe both firm-level financial policies and CEOs' personal financial portfolios over time. We draw on data from the Center for Corporate Governance Research for firm-level accounting, ownership, and governance information, and from Statistics Norway for comprehensive sociodemographic and financial information on individuals, including detailed data on personal wealth and asset holdings.

We focus on all limited liability firms between 2000 and 2013[1] and CEOs who served for at least one year during this period. Following prior research, we exclude subsidiaries, very

[1] The sample period is dictated by data availability. The integration of multiple sensitive datasets from different authorities requires substantial time and administrative coordination, limiting the feasibility of extending the data to more recent years.

small firms (fewer than five employees or assets below NOK 50 million), and regulated industries such as financials and utilities (Boustanifar, Zajac, and Zilja 2022; Zilja et al. 2023). We also require firms to have at least three annual observations. The final sample consists of 14,171 firm-year observations, 1,730 firms, and 3,094 CEOs. A key advantage of this setting is the availability of longitudinal, disaggregated data on CEOs' personal financial portfolios, which allows us to construct a revealed-preference measure of personal financial risk exposure and to track changes in that exposure over time.

Conceptual Mapping and Measurement Strategy

Our theoretical framework distinguishes between risk appetite (a latent target) and risk exposure (the realized allocation of risk across domains). Because risk appetite is not directly observable, our empirical approach focuses on measuring risk exposure in both personal and corporate domains. We operationalize personal risk exposure using the composition of CEOs' financial portfolios, and firm-level risk exposure using corporate financial policies that reflect financial risk-taking. In this sense, observed portfolio composition serves as a proxy for risk exposure, the observable counterpart to latent risk appetite. This mapping allows us to test both cross-sectional alignment (differences across CEOs) and intertemporal rebalancing (changes within CEOs over time).

We use two complementary measures of firm-level financial risk exposure: Leverage (total debt divided by total assets) and Cash ratio (cash and cash equivalents divided by total assets). Higher leverage increases financial risk exposure, while higher cash holdings reduce exposure. Consistent with prior literature, these measures capture key dimensions of firms' financial risk posture as they directly affect both downside exposure and financial flexibility (Rajan and Zingales 1995; Opler et al. 1999). In our analyses, we treat these as alternative

empirical indicators of the same underlying construct—firm-level financial risk exposure—rather than distinct theoretical outcomes.

For our independent variables, we first use Personal Financial Risk Exposure (Cross-Sectional Measure). To capture differences across CEOs, we measure personal financial risk exposure as the proportion of a CEO's financial wealth invested in risky assets, defined as stocks and equity mutual funds relative to total financial assets (including cash and bonds). This measure reflects the realized allocation of personal wealth to risky assets, which serves as an observable proxy for the CEO's target level of risk exposure. Unlike survey-based measures, it is based on actual financial decisions involving real stakes (Heaton and Lucas 2000).

Second, we use Changes in Personal Risk Exposure (Intertemporal Measure). To examine within-CEO adjustments over time, we construct a measure of changes in personal financial risk exposure. A central empirical challenge is to isolate variation in personal risk exposure not directly tied to contemporaneous firm decisions. To address this, we focus on changes in portfolio composition driven by asset value fluctuations rather than active trading. Specifically, we define ΔPortfolio Risk as the change in the proportion of risky assets from year t–1 to t in years when the CEO does not execute any trades. In these cases, changes in portfolio risk arise mechanically from changes in the relative value of risky versus safe assets. This approach allows us to capture variation in personal risk exposure that is less directly tied to active decision-making and more likely to reflect external shocks to asset values.

Clarification of Identification Assumptions

Our identification strategy exploits changes in CEOs' personal portfolio risk driven by asset price movements, which, conditional on controls and fixed effects, are less directly tied to contemporaneous firm-level financial policy choices. To strengthen this interpretation, we

include year fixed effects to absorb aggregate market movements and macroeconomic conditions, as well as industry fixed effects to control for industry-level shocks.

In additional analyses, we examine robustness to excluding or controlling for CEO holdings in the focal firm, thereby reducing concerns that portfolio changes are driven by firm-specific performance. While we do not claim that these shocks are fully exogenous, these steps help isolate variation in personal risk exposure that is not mechanically driven by contemporaneous firm decisions. Accordingly, our empirical strategy is best interpreted as identifying variation in personal risk exposure that is less directly tied to contemporaneous decision-making, rather than establishing strict exogeneity.

Other Variables

Our final independent variable of interest is a measure of CEO Power. To capture CEOs' ability to implement preferred allocations of risk exposure, we construct a composite measure of CEO power based on three commonly used indicators: (1) CEO–Chair duality (CEO also serves as board chair), (2) Low board independence (below-sample-average share of independent directors), and (3) Family CEO (CEO belongs to controlling family). These components capture different dimensions of CEO discretion, including structural authority, board oversight, and ownership-based influence. We sum these indicators to construct a CEO power index ranging from 0 to 3. This index captures the degree to which CEOs can influence firm-level decisions.

We also include a set of CEO- and firm-level controls commonly used in the literature on corporate financial policies and managerial decision-making (Cronqvist, Makhija, and Yonker 2012; Rajan and Zingales 1995; Zilja et al. 2023). CEO controls include age, tenure, education, wealth, compensation, ownership, and demographic characteristics. Firm controls include size, profitability, tangibility, age, ownership concentration, dividend status, and listing

status. To address concerns about potential overcontrol or collider bias, we report models with stepwise inclusion of controls and show that our results are robust across specifications.

Table 1 reports descriptive summary statistics for all the main variables used in our analyses. We report the correlation matrix and definition of variables in the Appendix Tables A1 and A2, respectively.

[INSERT TABLE 1ABOUT HERE]

Empirical Strategy

Our empirical analyses consist of three components. First, we conduct a cross-sectional analysis to test Hypothesis 1 (baseline alignment). Here, we estimate equation of the form:

$$\text{Firm Risk}_{ijt} = \beta_1 \text{Personal Risk}_j + \text{Controls} + \text{Industry FE} + \text{Year FE} \qquad (1)$$

where $i, j,$ and t stand for firm, CEO, and year, respectively. This specification captures whether CEOs with higher personal risk exposure are associated with higher firm-level financial risk-taking, controlling for observable characteristics and common shocks. Second, we conduct a Within-CEO (Intertemporal) Analysis to test Hypothesis 2 (rebalancing). Here, we estimate models with CEO–firm fixed effects:

$$\text{Firm Risk}_{ijt} = \beta_2 \Delta\text{Personal Risk}_{jt} + \text{Controls} + \text{CEO} - \text{Firm FE} + \text{Year FE} \qquad (2)$$

This specification isolates within-CEO–firm variation over time, controlling for all time-invariant characteristics of the CEO–firm match. The coefficient β_2 captures whether changes in personal risk exposure are associated with offsetting changes in firm-level risk-taking. Finally, we analyze Moderation by CEO Power to test Hypothesis 3. Here, we interact

both cross-sectional and intertemporal measures with CEO power. These models assess whether CEOs with greater discretion exhibit stronger alignment and rebalancing effects.

Additional Analyses and Robustness Checks

We conduct several additional analyses to strengthen identification and address alternative explanations. Specifically, we exclude CEO holdings in the focal firm to reduce concerns that changes in portfolio risk are mechanically driven by firm performance, estimate lagged specifications to better establish temporal ordering, and compare trading and non-trading CEOs to assess potential selection into the no-trade sample. We also examine robustness to the stepwise inclusion of controls to address concerns about overcontrol or collider bias. Together, these analyses provide additional support for interpreting our findings as consistent with the proposed mechanism.

RESULTS

We organize our results around the two core empirical implications of our framework. First, we examine cross-sectional alignment between CEOs' personal financial risk exposure and firm-level financial risk-taking. Second, we examine intertemporal rebalancing, testing whether changes in personal risk exposure within a CEO are associated with offsetting changes in firm-level risk-taking. Finally, we assess whether these relationships are moderated by CEO power.

Cross-Sectional Alignment in Risk Exposure

We begin by testing whether differences across CEOs in personal financial risk exposure are associated with differences in firm-level financial risk-taking. Table 2 reports the

results of regressions of firm leverage and cash holdings on CEOs' personal financial risk exposure, controlling for CEO and firm characteristics as well as industry and year fixed effects. Higher leverage reflects greater firm-level financial risk exposure, whereas higher cash holdings reflect lower financial risk exposure. Accordingly, the expected coefficient signs differ across the two dependent variables.

[INSERT TABLE 2 ABOUT HERE]

Consistent with our baseline prediction, we find that CEOs with higher personal financial risk exposure are associated with significantly higher firm leverage and lower cash holdings. These results indicate that CEOs who allocate a greater share of their personal wealth to risky assets also tend to pursue riskier corporate financial policies.

In terms of magnitude, an interquartile range increase in personal financial risk exposure is associated with a meaningful increase in firm leverage and a corresponding decrease in cash holdings. These findings are consistent with prior research documenting cross-sectional alignment between managerial characteristics and firm-level outcomes and provide a baseline validation of our empirical setting (Cronqvist et al., 2012).

Intertemporal Rebalancing in Risk Exposure

We next examine whether changes in personal financial risk exposure within a CEO are associated with compensatory changes in firm-level risk-taking. To do so, we estimate models with CEO–firm fixed effects, which isolate within-CEO variation over time.

Table 3 reports the results. Consistent with our rebalancing prediction, we find that increases in personal portfolio risk are associated with decreases in firm leverage and

increases in cash holdings. Conversely, decreases in personal risk exposure are associated with increases in firm-level financial risk-taking.

[INSERT TABLE 3 ABOUT HERE]

These findings indicate that when a CEO's personal portfolio becomes riskier, the firm's financial policies adjust in a direction that reduces corporate financial risk, and vice versa. This pattern is consistent with a process of *intertemporal rebalancing across domains*, in which CEOs adjust firm-level decisions in response to changes in their overall risk exposure. Importantly, these results are obtained using variation in personal portfolio risk driven by asset value changes rather than active trading, reducing concerns that the observed relationships are driven by simultaneous decision-making across domains. Consistent with this argument, we observe that CEOs in our sample infrequently adjust their personal portfolios through active trading: the median number of trades per year is zero, and the 75th percentile is only one trade per year. Taken together, these results provide substantial evidence consistent with the proposed rebalancing mechanism.

Moderating Role of CEO Power

We next examine whether the relationships documented above depend on CEOs' ability to influence firm-level decisions. Table 4 reports results from models interacting personal risk exposure and changes in exposure with our measure of CEO power. Consistent with our expectations, both the cross-sectional alignment and intertemporal rebalancing effects are stronger for CEOs with greater power relative to the board.

[INSERT TABLE 4 ABOUT HERE]

In the cross-sectional analysis, the positive relationship between personal and firm-level risk exposure is amplified when CEO power is higher. In the intertemporal analysis, the negative relationship between changes in personal risk exposure and firm-level risk-taking is also stronger for more powerful CEOs. These findings indicate that the ability to translate personal risk exposure into corporate policies depends on managerial discretion.

Additional Analyses and Robustness Checks

We conduct several additional analyses to assess the robustness of our findings and to address potential concerns regarding identification and alternative explanations. First, to mitigate concerns that changes in personal portfolio risk may be driven by firm-specific performance, we re-estimate our models excluding or controlling for CEOs' holdings in the focal firm. The results, not tabulated for brevity, remain qualitatively unchanged, in terms of similar magnitude and significance.

Second, we examine whether selection into the no-trade sample influences our findings by comparing CEOs who trade versus those who do not. The two groups appear to be very similar in terms of observable characteristics, based on the CEO-level control variables used in the paper, except that CEOs who trade tend to be slightly wealthier. We do not find significant differences between the two groups along other dimensions, such as tenure, education, CEO leverage, or CEO ownership in the firm. Third, we estimate models using lagged changes in personal risk exposure to strengthen causal interpretation. The results are consistent with our main findings, with a larger estimated effect in the lagged specifications. Finally, we assess the sensitivity of our results to the inclusion of control variables by estimating models with stepwise addition of controls. The key coefficients of

interest remain stable across specifications, alleviating concerns regarding overcontrol or collider bias.[2]

Taken together, our results provide consistent support for a cross-domain risk allocation perspective. We find evidence of both cross-sectional alignment between personal and firm-level risk exposure and intertemporal rebalancing within CEOs over time. These patterns are strongest when CEOs have greater discretion, consistent with the view that CEOs actively manage the allocation of risk across domains.

DISCUSSION

This study examined whether CEOs express risk preferences independently across domains or allocate risk across domains to maintain a target level of exposure. Our findings suggest that both patterns emerge, but that they are best understood as outcomes of a common underlying mechanism. Together, these findings indicate that CEO risk-taking is best understood as a cross-domain allocation process rather than a set of independent decisions.

Specifically, we show that differences across CEOs in personal financial risk exposure are associated with corresponding differences in firm-level financial risk-taking, consistent with prior research linking managerial characteristics to organizational outcomes. At the same time, we find that within a given CEO over time, increases in personal portfolio risk are associated with reductions in firm-level risk-taking, and vice versa. This pattern is consistent with a process of intertemporal rebalancing, in which CEOs adjust risk-taking across domains in response to changes in their overall exposure.

Taken together, these results suggest that personal and corporate decisions are jointly determined as part of a broader allocation of risk exposure. This perspective helps reconcile

[2] Appendix Table A3 reports the results for H1a; similar results are obtained for the other hypotheses.

what might otherwise appear to be conflicting empirical patterns—cross-sectional alignment and within-CEO offsetting adjustments—by showing that both can arise from a common logic of target risk allocation.

Our findings also highlight the importance of managerial discretion in implementing such allocation decisions. The effects we document are stronger when CEOs have greater power relative to the board, indicating that the ability to translate personal risk exposure into corporate financial policies depends on the degree of decision-making authority available to the CEO.

This study contributes to research on managerial risk-taking and corporate decision-making in several ways. First, we introduce a cross-domain portfolio perspective on CEO risk-taking. Prior research has largely examined how managerial characteristics influence decisions within the organizational domain, often treating those decisions as direct expressions of underlying preferences. In contrast, we show that CEOs simultaneously make risk-related decisions across multiple domains and that these decisions are interdependent. In doing so, we extend upper echelons theory by highlighting that executives' interpretations of their decision environment may span multiple domains, such that firm-level outcomes reflect not only trait expression within a domain but also the allocation of risk across domains. This perspective shifts attention from within-domain expression to cross-domain allocation, opening new avenues for examining how personal and professional decisions interact.

Second, we provide a unified framework that reconciles two seemingly opposing patterns in the relationship between CEO characteristics and firm behavior. Prior work has emphasized cross-sectional alignment between managerial traits and organizational outcomes. We show that this pattern can coexist with intertemporal rebalancing within CEOs, whereby changes in personal risk exposure lead to offsetting changes in firm-level risk-

taking. These two patterns, rather than representing competing explanations, reflect distinct manifestations of a common underlying mechanism of cross-domain risk allocation.

Third, we contribute by leveraging revealed-preference data on CEOs' personal financial portfolios to capture risk exposure. Prior research has often relied on indirect proxies or survey-based measures of managerial risk preferences (Dohmen et al., 2011). Using detailed administrative data on actual financial behavior, we provide a more direct and behaviorally grounded measure of personal risk exposure, enabling a sharper test of how individual-level risk exposure shapes firm-level financial policies.

While our empirical design does not allow us to fully rule out all alternative explanations or directly observe the decision processes underlying risk allocation, the convergence of cross-sectional alignment and within-CEO rebalancing patterns—derived from distinct sources of variation—provides consistent support for the proposed cross-domain allocation mechanism and is difficult to reconcile with explanations based solely on static preferences or purely within-domain mechanisms. Future research could build on this approach by examining the micro-level processes through which executives perceive and respond to changes in risk exposure.

In addition, although we focus on financial policies, the underlying logic of cross-domain risk allocation may extend to other domains of managerial decision-making, such as strategic investment, innovation, or career choices. Future work may also examine how constraints on adjustment—such as regulatory restrictions or governance structures—shape the extent to which CEOs rebalance risk across domains.

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Table 1: Descriptive Statistics

Variable	**Mean**	**SD**	**Median**
FIRM CHARACTERISTICS			
Leverage	0.41	0.23	0.39
Cash Ratio	0.12	0.15	0.06
Assets (logged)	19.22	1.37	18.81
Employees (logged)	3.87	1.40	3.83
Profitability	0.07	0.10	0.03
Firm Age (logged)	2.90	0.86	2.89
Listed	0.06	0.24	0.00
Dividend Paying	0.41	0.49	0.00
Ownership Concentration	0.63	0.37	0.58
Tangibility	0.24	0.28	0.11
Low Independent Board	0.48	0.50	0.00
CEO/Chair Duality	0.05	0.22	0.00
Family CEO	0.19	0.39	0.00
CEO Power Index	0.72	0.59	1
CEO CHARACTERISTICS			
CEO Risk Appetite	0.42	0.29	0.39
ΔPortfolio risk	0.00	0.39	-0.027
CEO Ownership	0.26	0.20	0.15
Leverage	2.23	4.65	1.20
Male	0.95	0.22	1.00
Married	0.68	0.47	1.00
Tenure	7.01	4.80	6.00
Age (logged)	3.89	0.17	3.91
Salary (logged)	14.26	0.78	14.13
Wealth (logged)	14.71	1.53	14.43
Education	5.53	1.40	6.00
Observations	14171		

Table 2: Behavioral Consistency: CEO risk appetite and firm policies

	(1)	(2)	(3)	(4)
	Baseline Alignment		Baseline Alignment	
Dependent Variable:	Firm Leverage		Firm Cash/Assets Ratio	
CEO Risk Appetite		**0.052**		**-0.038**
		(0.000)		**(0.000)**
CEO Male	-0.004	-0.005	-0.008	-0.007
	(0.642)	(0.554)	(0.184)	(0.233)
CEO Married	0.007	0.004	-0.004	-0.002
	(0.058)	(0.356)	(0.109)	(0.553)
CEO Tenure	0.000	-0.001	-0.001	-0.001
	(0.355)	(0.126)	(0.017)	(0.077)
CEO Age (logged)	-0.025	-0.019	0.049	0.045
	(0.034)	(0.110)	(0.000)	(0.000)
CEO Salary (logged)	-0.008	-0.006	0.012	0.011
	(0.014)	(0.035)	(0.000)	(0.000)
Financial Assets (logged)	-0.006	-0.007	-0.000	-0.000
	(0.000)	(0.000)	(0.392)	(0.973)
CEO Wealth (logged)	-0.007	-0.010	0.003	0.006
	(0.015)	(0.000)	(0.088)	(0.002)
CEO Education	-0.006	-0.006	0.004	0.005
	(0.000)	(0.000)	(0.000)	(0.000)
CEO International Background	-0.013	-0.013	-0.006	-0.006
	(0.040)	(0.041)	(0.195)	(0.191)
CEO Leverage	0.003	0.003	-0.001	-0.001
	(0.000)	(0.000)	(0.000)	(0.000)
Assets (logged)	0.055	0.056	-0.033	-0.034
	(0.000)	(0.000)	(0.000)	(0.000)
Employees (logged)	-0.025	-0.025	0.012	0.012
	(0.000)	(0.000)	(0.000)	(0.000)
Profitability	-0.279	-0.280	0.197	0.198
	(0.000)	(0.000)	(0.000)	(0.000)
Firm Age (logged)	-0.013	-0.013	0.001	0.001
	(0.000)	(0.000)	(0.556)	(0.427)
Listed	-0.053	-0.055	-0.007	-0.005
	(0.000)	(0.000)	(0.241)	(0.375)
Dividend Paying	-0.065	-0.065	0.012	0.012
	(0.000)	(0.000)	(0.000)	(0.000)
CEO Ownership	0.083	0.070	-0.056	-0.046
	(0.000)	(0.000)	(0.000)	(0.000)
Ownership Concentration	0.030	0.036	-0.004	-0.009
	(0.000)	(0.000)	(0.254)	(0.017)
Tangibility	0.168	0.170	-0.109	-0.110
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	14,171	14,171	14,171	14,171
R-squared	0.261	0.275	0.166	0.179

P-values are reported in parentheses. All regressions include industry and year fixed effects.

Table 3: Intertemporal Rebalancing: Changes in CEO Risky Portfolio and Firm Policies Within a CEO-Firm Match

	(1)	(2)	(3)	(4)
	Rebalancing		Rebalancing	
Dependent Variable:	Firm Leverage		Firm Cash/Assets Ratio	
ΔPortfolio risk		**-0.023**		**0.011**
		(0.005)		**(0.017)**
CEO Married	0.011	0.011	-0.006	-0.006
	(0.135)	(0.135)	(0.288)	(0.288)
CEO Tenure	0.002	0.002	0.002	0.002
	(0.366)	(0.367)	(0.259)	(0.259)
CEO Age (logged)	-0.355	-0.353	0.126	0.128
	(0.021)	(0.021)	(0.260)	(0.251)
CEO Salary (logged)	0.003	0.002	0.005	0.005
	(0.303)	(0.324)	(0.005)	(0.006)
Financial Assets (logged)	-0.001	-0.001	-0.002	-0.002
	(0.600)	(0.659)	(0.064)	(0.090)
CEO Wealth (logged)	-0.013	-0.013	0.004	0.005
	(0.000)	(0.000)	(0.065)	(0.055)
CEO Education	0.041	0.041	-0.029	-0.030
	(0.095)	(0.097)	(0.215)	(0.212)
CEO Leverage	0.001	0.001	0.000	0.000
	(0.010)	(0.011)	(0.845)	(0.843)
Assets (logged)	0.077	0.077	-0.001	-0.001
	(0.000)	(0.000)	(0.827)	(0.829)
Employees (logged)	-0.016	-0.017	-0.000	-0.000
	0.000	(0.000)	(0.886)	(0.876)
Profitability	-0.227	-0.227	0.120	0.120
	(0.000)	(0.000)	(0.000)	(0.000)
Firm Age (logged)	-0.005	-0.005	-0.025	-0.025
	(0.544)	(0.547)	(0.000)	(0.000)
Dividend Paying	-0.049	-0.048	0.012	0.012
	(0.000)	(0.000)	(0.000)	(0.000)
CEO Ownership	0.015	0.016	-0.023	-0.022
	(0.336)	(0.324)	(0.047)	(0.051)
Ownership Concentration	0.027	0.027	-0.007	-0.007
	(0.010)	(0.009)	(0.414)	(0.399)
Tangibility	0.155	0.155	-0.193	-0.193
	(0.000)	(0.000)	(0.000)	(0.000)
Observations	14,171	14,171	14,171	14,171
R-squared	0.848	0.849	0.785	0.785

P-values are reported in parentheses. All regressions include CEO-Firm, industry, and year fixed effects.

Table 4: The moderating role of CEO power

	(1)	(2)	(3)	(4)
	CEO Power Moderation			
Dependent Variable:	Firm Leverage		Firm Cash/Assets Ratio	
CEO Power Index				
× CEO Risk Appetite	**0.024**		**-0.019**	
	(0.000)		**(0.000)**	
CEO Power Index				
× ΔPortfolio risk		**-0.009**		**0.003**
		(0.002)		**(0.033)**
Average effect of CEO Risk Appetite	0.052	-0.023	-0.038	0.011
Controls	Yes	Yes	Yes	Yes
Moderating variables separately	Yes	Yes	Yes	Yes
Observations	14,171	14,171	14,171	14,171
R-squared	0.275	0.860	0.183	0.790

The table reports the estimated coefficients and p-values of our main variables of interest while testing H2 (CEO power moderating the relation in H1a and H1b) and H4 (CEO power moderating the relation in H3a and H3b). Our variables of interest are two interaction variables: *CEO Power Index* x *CEO Risk Appetite* in H2, *CEO Power Index* x *ΔPortfolio risk* in H4.

The complete specification of regressions is provided in Equation (3) in the paper. We note that each column includes all control variables as well as the interaction variables as separate regressors. For example, in column 1, CEO Power Index and CEO Risk Appetite are also separately included along with control variables in the regression, but for the sake of brevity, their estimated coefficients are not reported.

Appendix Tables

Table A1: Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)
(1) Firm Leverage	1																								
(2) Firm Cash Ratio	-0.26	1																							
(3) CEO Risk Propensity	-0.02	-0.04	1																						
(4) Δportfolio risk	-0.02	-0.03	0.29	1																					
(5) Assets (logged)	0.19	-0.18	-0.05	-0.04	1																				
(6) Employees (logged)	-0.08	0.06	-0.16	-0.14	0.32	1																			
(7) Profitability	-0.16	0.13	0.10	0.06	-0.05	-0.03	1																		
(8) Firm Age (logged)	-0.10	-0.01	0.09	0.05	0.06	0.06	0.01	1																	
(9) Listed	-0.01	-0.04	0.12	0.12	0.35	0.05	-0.03	0.11	1																
(10) Dividend Paying	-0.19	0.03	0.10	0.09	0.06	-0.02	0.23	0.14	-0.01	1															
(11) CEO Ownership	0.01	-0.04	0.49	0.11	-0.19	-0.19	0.06	0.02	-0.09	0.07	1														
(12) Ownership Concentration	0.05	0.01	-0.36	-0.02	0.01	0.23	-0.03	-0.09	-0.27	-0.17	-0.09	1													
(13) Tangibility	0.22	-0.20	-0.09	-0.08	-0.10	-0.02	-0.06	0.00	-0.16	-0.01	0.01	-0.01	1												
(14) CEO Male	-0.02	-0.03	0.08	0.07	0.00	-0.04	0.01	0.02	0.03	0.04	0.03	-0.06	0.00	1											
(15) CEO Married	0.02	-0.02	0.21	0.16	0.11	0.01	0.02	0.04	0.08	0.00	0.03	-0.08	-0.04	0.12	1										
(16) CEO Tenure	-0.09	0.00	0.28	0.22	-0.14	-0.13	0.03	0.27	-0.07	0.14	0.27	-0.13	0.04	0.10	0.02	1									
(17) CEO Age (logged)	-0.04	0.03	0.06	0.03	0.08	0.03	-0.01	0.11	-0.02	0.09	0.04	-0.01	0.07	0.11	0.19	0.34	1								
(18) CEO Salary (logged)	-0.06	0.05	0.27	0.20	0.36	0.12	0.10	0.08	0.23	0.07	0.17	-0.06	-0.28	0.03	0.13	0.11	0.11	1							
(19) CEO Wealth (logged)	-0.08	0.02	0.65	0.57	0.08	-0.12	0.06	0.11	0.12	0.10	0.53	-0.26	-0.16	0.07	0.15	0.32	0.19	0.53	1						
(20) CEO Education	-0.03	0.04	-0.08	-0.07	0.24	0.14	-0.04	-0.03	0.11	0.00	-0.20	0.00	-0.08	-0.03	0.03	-0.15	-0.04	0.07	-0.07	1					
(21) CEO International Background	-0.03	0.01	-0.03	-0.02	0.00	0.06	-0.01	0.00	-0.01	-0.04	0.00	0.10	-0.07	-0.02	-0.02	-0.03	-0.06	0.06	-0.01	0.04	1				
(22) CEO Leverage	0.09	-0.06	0.19	0.18	-0.05	-0.14	-0.01	-0.01	0.00	-0.04	0.26	-0.07	0.00	0.03	0.00	0.06	-0.07	0.04	0.24	-0.07	0.03	1			
(23) CEO/Chair Duality	0.01	-0.01	0.19	0.15	-0.10	-0.09	0.03	0.00	-0.06	0.00	0.40	0.02	0.01	0.04	0.04	0.10	0.07	0.06	0.22	-0.12	-0.01	0.11	1		
(24) Family CEO	0.01	-0.04	0.15	0.05	-0.19	-0.22	0.06	0.05	-0.07	0.09	0.72	-0.28	0.03	0.01	0.01	0.28	0.00	0.14	0.54	-0.21	-0.03	0.26	0.30	1	
(25) Low Independent Board	0.03	0.03	-0.24	-0.20	0.14	0.00	-0.16	-0.03	0.10	-0.04	-0.28	0.05	0.12	0.00	-0.04	-0.09	0.05	-0.08	-0.21	0.13	-0.02	-0.07	-0.18	-0.31	1

Table A2: List of Variables with Definitions

Variable Name	Definition
FIRM CHARACTERISTICS	
Leverage	D/(D+E) where D is debt and E is equity.
Cash Ratio	(Cash and Cash Equivalents)/Total Assets
Assets (logged)	log of total assets
Employees (logged)	log of number of employees
Profitability	EBIT (1-t)/Assets
Firm Age (logged)	log of firm age
Listed	1 if the firm is listed in a stock exchange and 0 otherwise
Dividend Paying	1 if the firm is paying dividend and 0 otherwise
Ownership Concentration	Herfindahl index based on ownership concentration. It is the sum of the squares of the ownership of the largest 5 shareholders in the firm. It could range between 0 (very dispersed ownership) to 1 (only one owner)
Tangibility	The ratio tangible to total assets
Low Independent Board	1 if the share of independent board members below the median in the sample and 0 otherwise
CEO/Chair Duality	1 if the CEO is also the chairman of the board and 0 otherwise
Family CEO	1 if the CEO is from the largest family owner and 0 otherwise
CEO Power Index	The sum of the following dummy variables: low independent board, Family CEO, and CEO-Chair duality. Therefore it could range between 0 and 3.
CEO CHARACTERISTICS	
CEO Risk Propensity	The average share of CEO financial assets invested in risky securities (stocks and equity funds)
ΔPortfolio risk	Passive changes in riskiness of CEO portfolio where riskiness is defined as the proportion of financial assets invested in stocks and funds.
CEO Ownership	Percentage of CEO ownership in the firm
Leverage	CEO total debt to income ratio
Male	1 if the CEO is male and 0 otherwise
Married	1 if the CEO is married and 0 otherwise
Tenure	The number of years the CEO has been at the firm
Age (logged)	log of the CEO age
Salary (logged)	log of the CEO salary
Wealth (logged)	log of CEO taxable wealth
Education	8 categories based on the number of education years. **0:** No education. **1:** 1-7 years; **2:** 8-10 years; **3:** 11-12 years; **4:** 13 years; **5:** 14 years; **6:** 14-17 years; **7:** 18-19 years; **8:** 20 years and more
International Background	1 if the either the CEO or one of her parents are born abroad

Table A3: Stepwise inclusion of control variables

Dep. Var:	Firm leverage				
CEO Risk Appetite	**0.054**	**0.050**	**0.055**	**0.051**	**0.052**
	(0.000)	**(0.009)**	**(0.003)**	**(0.000)**	**(0.000)**
CEO Male		0.008	0.005	-0.003	-0.005
		(0.338)	(0.526)	(0.759)	(0.554)
CEO Married		0.009	0.009	0.002	0.004
		(0.041)	(0.026)	(0.535)	(0.356)
CEO Tenure		-0.003	-0.003	-0.001	-0.001
		(0.000)	(0.000)	(0.135)	(0.126)
CEO Age (logged)		-0.010	0.004	-0.006	-0.019
		(0.436)	(0.752)	(0.634)	(0.110)
CEO Salary (logged)		-0.004	0.000	-0.017	-0.006
		(0.144)	(0.874)	(0.000)	(0.035)
Financial Assets (logged)		-0.011	-0.008	-0.008	-0.007
		(0.000)	(0.000)	(0.000)	(0.000)
CEO Wealth (logged)			-0.008	-0.011	-0.010
			(0.010)	(0.000)	(0.000)
CEO Education			-0.004	-0.008	-0.006
			(0.006)	(0.000)	(0.000)
CEO International Background			-0.015	-0.018	-0.013
			(0.022)	(0.005)	(0.041)
CEO Leverage			0.005	0.003	0.003
			(0.000)	(0.000)	(0.000)
Assets (logged)				0.048	0.056
				(0.000)	(0.000)
Employees (logged)				-0.021	-0.025
				(0.000)	(0.000)
Profitability				-0.301	-0.280
				(0.000)	(0.000)
Dividend Paying				-0.067	-0.065
				(0.000)	(0.000)
CEO Ownership				0.090	0.070
				(0.000)	(0.000)
Ownership Concentration				0.045	0.036
				(0.000)	(0.000)
Firm Age (logged)					-0.013
					(0.000)
Listed					-0.055
					(0.000)
Tangibility					0.170
					(0.000)
Observations	14,171	14,171	14,171	14,171	14,171
R-squared	0.140	0.151	0.159	0.244	0.275